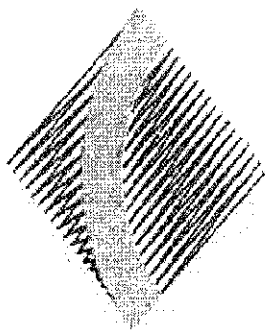


JG
AW
AT

Name: _____
Class: 12MTX _____
Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2008 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 1

*Time allowed - 2 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 7.

*****Each page must show your name and your class. *****

	Marks
QUESTION ONE	
a) Find the integral $\int \sec^2 2x \, dx$.	1
b) Evaluate the limit $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{5x}{2}\right)}{3x}$	2
c) Find the exact value of $\sin 15^\circ$.	2
d) Find the values of x for which $\frac{x}{x+1} \geq 2$.	3
e) Find the point that divides the line joining A (2, 8) and B (-5, 7) in the ratio 2 : 3 externally.	2
f) The line L makes an angle of 45° with the line $x - 2y + 3 = 0$. Find the gradient, m , of line L given that $m > 0$.	2

QUESTION TWO (START A NEW PAGE)

a) (i) Prove that $\cot\left(\frac{\alpha}{2}\right) = \frac{\sin \alpha}{1 - \cos \alpha}$.	2
(ii) Hence find the value of $\cot\left(\frac{\alpha}{2}\right)$ when $\sin \alpha = \frac{3}{5}$ and $\frac{\pi}{2} < \alpha < 2\pi$.	2
b) (i) Show that there is a real root of the equation $2 \tan x + 2x - \pi = 0$ between $x = 0.6$ and $x = 0.75$.	1
(ii) Start with $x = 0.6$, and use one application of Newton's method to approximate the root of $2 \tan x + 2x - \pi = 0$ in part (i). Give your answer correct to 2 decimal places.	2
c) $(2 + 5x)^n$ is expanded in ascending powers of x .	
(i) Write down the coefficient of the 8 th term in the expansion.	1
(ii) Show that $\frac{\text{the coefficient of the 8th term}}{\text{the coefficient of the 10th term}} = \frac{288}{25(n-7)(n-8)}$.	2
(iii) Hence determine the value of n if $\frac{\text{the coefficient of the 8th term}}{\text{the coefficient of the 10th term}} = \frac{36}{175}$.	2

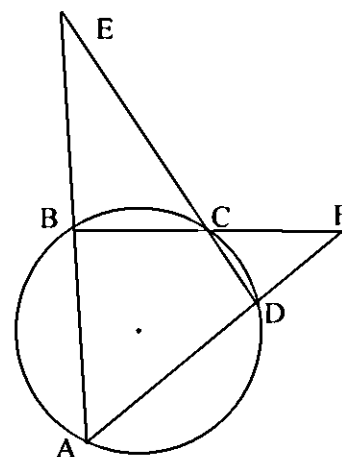
QUESTION THREE (START A NEW PAGE)

- a) Find the exact value of $\int_0^{\sqrt[4]{3}} \frac{dx}{16+x^2}$. 2
- b) Find the general solution to the equation $6 \sin^2 x + 5 \sin x - 4 = 0$. 2
- c) (i) Use the substitution $u = 1 - x^2$ to find the integral $\int \frac{x}{\sqrt{1-x^2}} dx$. 2
- (ii) Find the derivative of $x \sin^{-1} x$. 1
- (iii) Hence find the integral $\int \sin^{-1} x dx$. 2
- d) Given $f(x) = 3 \sin^{-1}(4x - 1)$,
 (i) find the domain and range of $f(x)$. 2
 (ii) Sketch the graph of $y = f(x)$. 1

QUESTION FOUR (START A NEW PAGE)

- a) In the diagram shown on the right, ABE, BCF, ADF and ECD are all straight lines and $\angle AED = \angle BFD$.

- (i) Explain why $\angle ABC = \angle ADC$. 2
 (ii) Hence prove that AC is a diameter. 1



- b) Prove, by mathematical induction, that

$$11 \times 2! + 19 \times 3! + 29 \times 4! + \dots + (n^2 + 5n + 5)(n+1)! = (n+4)[(n+2)!] - 8 \quad 3$$

- c) Show that if $x = \alpha$ is a double root of the equation $P(x) = 0$, then $x = \alpha$ is also a root of the equation $P'(x) = 0$. 2
- d) $P(x) = kx^4 - (2k+5)x^3 + (2k+10)x^2 - (2k+5)x + k$, where k is an integer.
 (i) Show that $x = 1$ is a double root of $P(x) = 0$. [You can assume the result of part (c)] 1
 (ii) Show that if $x = \alpha$ ($\alpha \neq 1$) is a root of $P(x) = 0$, then $x = \frac{1}{\alpha}$ is another root of $P(x) = 0$. 2
 (iv) Hence, show that $\alpha^2 + \frac{1}{\alpha^2} = \frac{25}{k^2} - 2$. 1

QUESTION FIVE (START A NEW PAGE)

Marks

- a) The velocity, $v \text{ ms}^{-1}$, of a particle travelling along a straight line is given by the expression $v^2 = 48 + 16x - 4x^2$.

(i) Show that the particle executes simple harmonic motion.

2

(ii) Find the amplitude of the motion.

1

(iii) If the particle starts from the point furthest to the right, find the expression for its position x , in terms of time t .

2

- b) Robin Hood shot an arrow with a speed of $V \text{ ms}^{-1}$ at an angle of 60° to the horizontal.

(i) Derive the expression for the vertical component of the displacement of the arrow at time t . [Ignore air resistance and you may assume the result $\dot{y} = -gt + \frac{V \sin 60^\circ}{1}$]

1

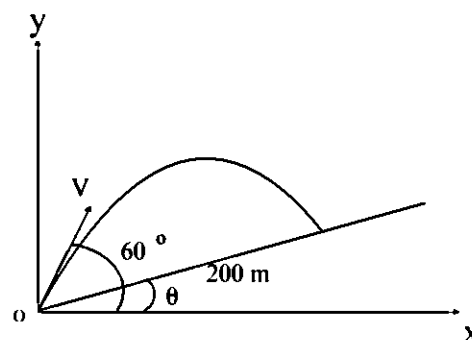
(ii) Show that the Cartesian equation of the path of the arrow is given by $y = \sqrt{3}x - \frac{2gx^2}{V^2}$.

1

[You may assume $x = \frac{Vt \cos 60^\circ}{1}$.]

(iii) Robin Hood stood at the bottom of a hill inclined at an angle θ to the horizontal and shot an arrow at an angle of 60° to the horizontal at a speed of $V \text{ ms}^{-1}$. He could shoot 200m up the hill. Use the result of part (ii) to show that

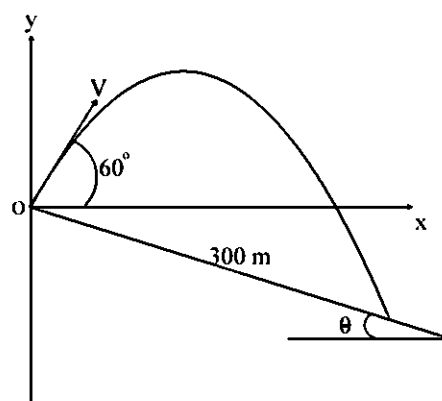
$$\tan \theta = \sqrt{3} - \frac{400g \cos \theta}{V^2}.$$



2

(iv) If he was standing on the hill and shot an arrow at the same speed of $V \text{ ms}^{-1}$ and the same angle of projection of 60° , but down the hill, he could shoot 300m down the hill. Show that

$$\tan \theta = \frac{600g \cos \theta}{V^2} - \sqrt{3}.$$



1

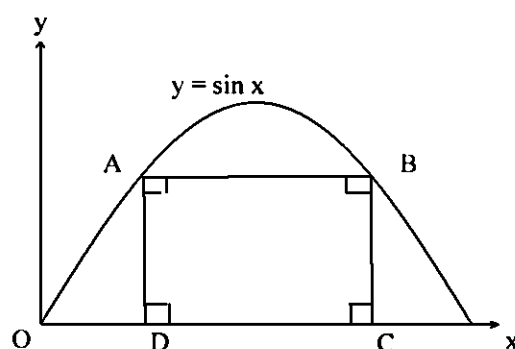
v) Hence, show that $\theta = \tan^{-1}\left(\frac{\sqrt{3}}{5}\right)$.

2

QUESTIONS SIX (START A NEW PAGE)

Marks

a)



The diagram shows a rectangle inscribed under the curve $y = \sin x$ in $0 \leq x \leq \pi$.

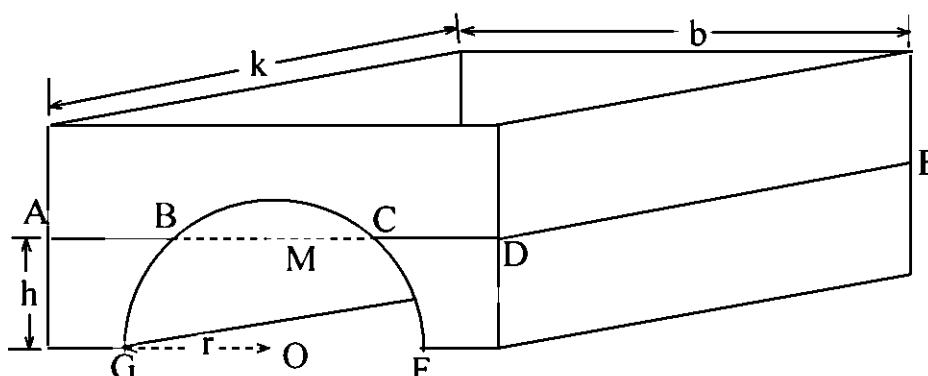
- (i) The coordinates of point A are $(x, \sin x)$. Explain why the coordinates of B are $(\pi - x, \sin x)$. 1
- (ii) Show that the area $A(x)$ of the rectangle ABCD is given by $A(x) = (\pi - 2x)\sin x$. 1
- (iii) Hence determine the dimensions of the rectangle with the largest area that can be inscribed under the graph $y = \sin x$, $0 \leq x \leq \pi$. [You may assume the result in part (b) (ii) of Question Two.]. 3

b) Given $f(x) = \sin^{-1}\left(\frac{1}{x}\right) + \cos^{-1}\left(\frac{1}{x}\right)$.

- (i) Find the derivative of $f(x)$. 2
- (ii) Find the domain of $f(x)$ and sketch the graph of $y = f(x)$. 3
- (iii) Find the coordinates of the point(s) of intersection between $y = f(x)$ and $y = f^{-1}(x)$. 2

Question Seven is on the next page....

a)



The diagram above shows a water trough in the shape of a rectangular prism with a half cylindrical cavity, of radius r , at the bottom. The length and width of the trough are k and b respectively. It is partly filled with water to a depth of h ($h < r$). Let O be the centre of the semi-circle $BCFG$ and M is the mid-point of BC . AB , CD and DE show the water surface inside the trough.

- (i) Express BM in terms of r and h . 1
 - (ii) Hence show that the area A of the water surface is given by $A = k \left[b - 2\sqrt{r^2 - h^2} \right]$. 1
 - (iii) The water in the trough is evaporating in such a way that h is decreasing at a constant rate. The trough measurements are $k = 3$ m, $b = 2$ m, $r = 50$ cm, and the water surface is descending at a constant rate of 0.6 cm/day. Find the rate at which the surface area is decreasing when the depth of the water in the trough is 30 cm. 3
- b) The variable point P has coordinates $(a \cos 2\theta, a \cos \theta)$.
- (i) Show that P lies on the curve $y^2 = \frac{a}{2}(x + a)$. 2
 - (ii) Sketch the locus of P as θ varies, taking account of any restriction on x and y . Label the focus and the vertex. 3
 - (iii) Find the equation of the tangent to the curve at the point where $\theta = \frac{\pi}{3}$. 2

E N D

Solution to Ext 1 AP4 2008

Question 1

$$a) \int \sec^2 x dx = \frac{1}{2} \tan 2x + C$$

✓

$$b) \lim_{x \rightarrow 0} \frac{\sin(\frac{5x}{2})}{3x} = \lim_{x \rightarrow 0} \frac{5}{6} \cdot \frac{\sin \frac{5x}{2}}{\frac{5x}{2}}$$

✓

$$= \frac{5}{6}$$

✓

$$c) \sin 15^\circ = \sin(45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ - \sin 30^\circ \cos 45^\circ$$

✓

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

✓

Alt 1

$$\sin 15^\circ = \sin(60^\circ - 45^\circ)$$

$$= \sin 60^\circ \cos 45^\circ - \sin 45^\circ \cos 60^\circ$$

✓

$$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

✓

Alt 2

$$2 \sin^2 15^\circ = 1 - \cos 30^\circ$$

$$= 1 - \frac{\sqrt{3}}{2}$$

$$\therefore \sin^2 15^\circ = \frac{2 - \sqrt{3}}{4}$$

✓

$$\sin 15^\circ = \sqrt{\frac{2 - \sqrt{3}}{4}}, \quad -\sqrt{\frac{2 - \sqrt{3}}{4}} \text{ (rejected, since } \sin 15^\circ > 0)$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

✓

$$d) \frac{x}{x+1} \geq 2$$

$$x(x+1) \geq 2(x+1)^2$$

✓

$$2(x+1)^2 - x(x+1) \leq 0$$

$$(x+1)[2(x+1) - x] \leq 0$$

$$(x+1)(x+2) \leq 0$$

✓

$$-2 \leq x \leq -1$$

✓

[Both inequality signs must be correct]

e) At the point of division,

$$x = \frac{(-2)(-5) + (3)(2)}{-2 + 3}$$

$$= 16$$

✓

$$y = \frac{(-2)(7) + (3)(8)}{-2 + 3}$$

$$= 10$$

✓

$\therefore$ The point is (16, 10)

f)

$$x - 2y + 3 = 0$$

$$y = \frac{1}{2}(x+3)$$

$$m_1 = \frac{1}{2}$$

$$\tan 45^\circ = \left| \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} \right|$$

✓

$$\therefore \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} = 1 \quad \text{or} \quad \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} = -1$$

$$\frac{2m-1}{2+m} = 1 \quad \frac{2m-1}{2+m} = -1$$

$$2m-1 = 2+m \quad 2m-1 = -2-m$$

$$m = 3 \quad \checkmark$$

$$3m = -1$$

$$m = -\frac{1}{3} \text{ (rejected, } m > 0 \text{)}$$

Question 2

(a) (i) Let $t = \tan \frac{\alpha}{2}$

$$\text{then } \frac{\sin \alpha}{1 - \cos \alpha} = \frac{\frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}} \quad \checkmark$$

$$= \frac{2t}{1+t^2 - (1-t^2)}$$

$$= \frac{2t}{2t^2}$$

$$= \frac{1}{t} \quad \checkmark$$

$$= \frac{1}{\tan \frac{\alpha}{2}}$$

$$= \cot \frac{\alpha}{2}$$

(ii) since $\sin \alpha > 0$ and $\frac{\pi}{2} < \alpha < 2\pi$

$$\therefore \frac{\pi}{2} < \alpha < \pi$$

$$\text{hence } \frac{\pi}{4} < \frac{\alpha}{2} < \frac{\pi}{2}$$

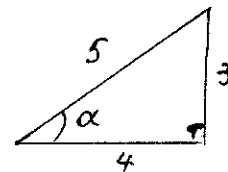
$$\text{then } \cos \alpha < 0 \text{ and } \cot \frac{\alpha}{2} > 0$$

$$\cos \alpha = -\frac{4}{5} \quad \checkmark$$

$$[\text{or } \cos \alpha = \sqrt{1 - \sin^2 \alpha}]$$

$$= \sqrt{1 - \left(\frac{3}{5}\right)^2}$$

$$= -\frac{4}{5} \quad \text{since } \cos \alpha < 0 \quad]$$



By result of (i)

$$\cot \frac{\alpha}{2} = \frac{\sin \alpha}{1 - \cos \alpha}$$

$$= \frac{\frac{3}{5}}{1 - \left(-\frac{4}{5}\right)}$$

$$= \frac{3}{5+4}$$

$$= \frac{1}{3} \quad \checkmark$$

b) (i) Let $f(x) = 2 \tan x + 2x - \pi$

$$f(0.6) = -0.5733 < 0$$

$$f(0.75) = 0.2216 > 0$$

since $f(0.6)$ and $f(0.75)$ are opposite in sign, $\checkmark$

there must be a root of $f(x) = 0$ between $x = 0.6$

and $x = 0.75$

(ii) $f'(x) = 2 \sec^2 x + 2$

With one application of Newton's method, the next approximation is given by

$$x = 0.6 - \frac{f(0.6)}{f'(0.6)} \quad \boxed{\checkmark}$$

$$= 0.6 - \frac{-0.5733}{4.9361}$$

$$= 0.71614$$

$$= 0.72 \text{ (2 dpl.)} \quad \boxed{\checkmark}$$

$$c) (i) (2+5x)^n = \sum_{r=0}^n \binom{n}{r} 2^{n-r} (5x)^r$$

$$\therefore \text{coeff. of the } 8^{\text{th}} \text{ term} = \binom{n}{7} 2^{n-7} 5^7 \quad \boxed{\checkmark}$$

$$(ii) \text{coeff of the } 10^{\text{th}} \text{ term} = \binom{n}{9} 2^{n-9} 5^9$$

$$\frac{\text{coeff. of the } 8^{\text{th}} \text{ term}}{\text{coeff of the } 10^{\text{th}} \text{ term}} = \frac{\binom{n}{7} 2^{n-7} 5^7}{\binom{n}{9} 2^{n-9} 5^9} \quad \boxed{\checkmark}$$

$$= \frac{n!}{7!(n-7)!} \times \frac{9!(n-9)!}{n!} \times \frac{2^2}{5^2}$$

$$= \frac{9 \times 8}{(n-7)(n-8)} \times \frac{4}{25} \quad \boxed{\checkmark}$$

$$= \frac{288}{25(n-7)(n-8)}$$

$$(iii) \frac{\text{coeff. of the } 8^{\text{th}} \text{ term}}{\text{coeff. of the } 10^{\text{th}} \text{ term}} = \frac{26}{175}$$

$$\frac{288}{25(n-7)(n-8)} = \frac{26}{175}$$

$$\frac{8}{(n-7)(n-8)} = \frac{1}{7} \quad \boxed{\checkmark}$$

$$56 = (n-7)(n-8)$$

$$56 = n^2 - 15n + 56$$

$$n^2 - 15n = 0$$

$$n(n-15) = 0$$

$$\therefore n = 0 \text{ (rejected), } n = 15. \quad \boxed{\checkmark}$$

Question 3

$$a) \int_0^{\sqrt{3}} \frac{dx}{16+x^2} = \frac{1}{4} \left[\tan^{-1} \frac{x}{4} \right]_0^{\sqrt{3}} \quad \boxed{\checkmark}$$

$$= \frac{1}{4} \left[\tan^{-1} \sqrt{3} - 0 \right]$$

$$= \frac{1}{4} \cdot \frac{\pi}{3}$$

$$= \frac{\pi}{12} \quad \boxed{\checkmark}$$

$$b) 6 \sin^2 x + 5 \sin x - 4 = 0$$

$$(2 \sin x - 1)(3 \sin x + 4) = 0$$

$$\therefore \sin x = \frac{1}{2} \quad \text{or} \quad \sin x = -\frac{4}{3} \text{ (rejected)}$$

$$x = \frac{\pi}{6}, \pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}, 3\pi - \frac{\pi}{6}, \dots$$

$$= n\pi + (-1)^n \frac{\pi}{6}$$

(c) (i)

$$u = 1 - x^2$$

$$\frac{du}{dx} = -2x$$

$$\therefore x dx = -\frac{1}{2} du$$

$$\int \frac{x dx}{\sqrt{1-x^2}} = -\int \frac{\frac{1}{2} du}{\sqrt{u}}$$

$$= -\sqrt{u} + C$$

$$= -\sqrt{1-x^2} + C$$

$$(ii) \frac{d}{dx} x \sin^{-1} x = \sin^{-1} x + \frac{x}{\sqrt{1-x^2}}$$

$$(iii) \int \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} dx = x \sin^{-1} x + C$$

$$\int \sin^{-1} x dx + \int \frac{x}{\sqrt{1-x^2}} dx = x \sin^{-1} x + C$$

$$\int \sin^{-1} x dx - \sqrt{1-x^2} = x \sin^{-1} x + C$$

$$\int \sin^{-1} x dx = x \sin^{-1} x + \sqrt{1-x^2} + C$$

$$d) (i) \text{ domain: } -1 \leq 4x-1 \leq 1$$

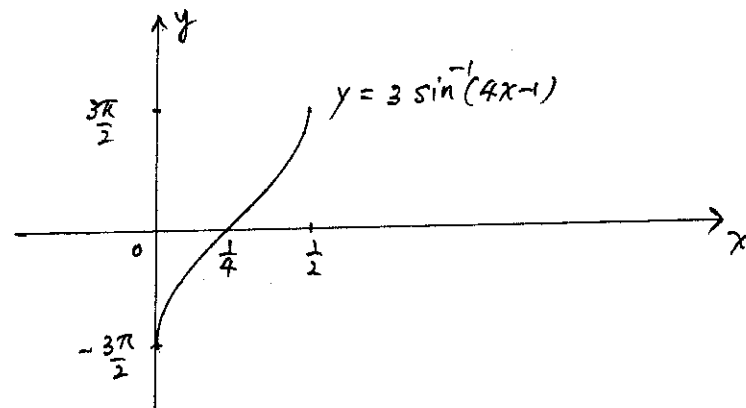
$$0 \leq 4x \leq 2$$

$$0 \leq x \leq \frac{1}{2}$$

$$\text{range: } -\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

(ii)



Question 4

a) In $\triangle ABF$ and $\triangle ADE$

$$\angle BAF = \angle DAE$$

(common)

$$\angle BFA = \angle DEA$$

(given)

$$\therefore \triangle ABF \sim \triangle ADE$$

(equiangular $\triangle$ s)

$$\therefore \angle ABC = \angle ADC$$

(Corresponding $\angle$ s of similar $\triangle$ s)

Alt

$$\angle BCE = \angle DCE$$

(vert. opp. $\angle$ s)

$$\angle BEC = \angle CED$$

(given)

$$\angle ABC = \angle BEC + \angle BCE$$

(ext $\angle$ of $\triangle$ equals int. opp $\angle$'s sum)

$$= \angle CED + \angle DCE$$

(proved)

$$= \angle ADC$$

(ext $\angle$ of $\triangle$)

(ii) $\angle ABC + \angle ADC = 180^\circ$ (opp $\angle$ s of cyclic quad are supplementary)

$\therefore 2\angle ABC = 180^\circ$ ($\angle ABC = \angle ADC$, proven)

$\angle ABC = 90^\circ$ ✓

$\therefore AC$ is a diameter ($\angle$ in semi-circle is a rt $\angle$.)

b) When $n=1$,

$11 \times 2! = 22$

$(n+4)(n+2)! - 8 = 5 \times 3! - 8$
 $= 22$

$\therefore$ It is true for $n=1$ ✓

Assume it is true for k , where k is an integer;

i.e. $11 \times 2! + 19 \times 3! + 29 \times 4! + \dots + (k^2 + 5k + 5)(k+1)!$
 $= (k+4)(k+2)! - 8$

then $11 \times 2! + 19 \times 3! + 29 \times 4! + \dots + (k^2 + 5k + 5)(k+1)!$
 $+ [(k+1)^2 + 5(k+1) + 5](k+2)!$

$= (k+4)(k+2)! - 8 + (k^2 + 7k + 11)(k+2)!$ by assumption. ✓

$= (k+2)! [(k+4) + (k^2 + 7k + 11)] - 8$

$= (k+2)! (k^2 + 8k + 15) - 8$

$= (k+2)! (k+3)(k+5)$

$= (k+5)(k+3)! - 8$

$= [(k+1)+4][(k+1)+2]! - 8$ } ✓

$\therefore$ It will be true for $n=k+1$ if it is true

for $n=k$. Since it is proved true for $n=1$,
 $\therefore$ it will be true for $n=2, 3, 4, \dots$ all integers n .

(c) Since $x=\alpha$ is a double root of $P(x)=0$

$\therefore P(x) = (x-\alpha)^2 Q(x)$ ✓

where $Q(x)$ is a polynomial in x .

$P'(x) = 2(x-\alpha)Q(x) + (x-\alpha)^2 Q'(x)$

$= (x-\alpha)[2Q(x) + (x-\alpha)Q'(x)]$ ✓

$\therefore x-\alpha$ is a factor of $P'(x)$

i.e. $x=\alpha$ is a root of $P'(x)=0$

d) (i) $P(x) = kx^4 - (2k+5)x^3 + (2k+10)x^2 - (2k+5)x + k$

$P'(x) = 4kx^3 - 3(2k+5)x^2 + 2(2k+10)x - (2k+5)$

$P'(1) = 4k - 3(2k+5) + 2(2k+10) - (2k+5)$

$= 0$

$\therefore x=1$ is a root of $P'(x)=0$, hence $x=1$ ✓
 is a double root of $P(x)=0$ by part (c).

(ii) Since $x=1$ is a double root and $x=\alpha$ is a root, $\therefore$ let the 4th root be β .

product of roots $(1)(1)\alpha\beta = \frac{k}{k}$ ✓

$\alpha\beta = 1$

$$\beta = \frac{1}{\alpha} \quad \left. \begin{array}{l} \therefore x = \frac{1}{\alpha} \text{ is another root of } P(x)=0 \end{array} \right\} \boxed{\checkmark}$$

(iv) Sum of squares of roots

$$1^2 + 1^2 + \alpha^2 + \frac{1}{\alpha^2} = (\text{sum of roots})^2 - 2 \times \text{sum of roots taken two at a time}$$

$$= \left(\frac{2k-5}{k} \right)^2 - \frac{2(2k+10)}{k}$$

$$= \frac{4k^2 - 20k + 25 - 4k^2 - 20k}{k^2}$$

$$= \frac{25}{k^2} \quad \boxed{\checkmark}$$

$$\therefore \alpha^2 + \frac{1}{\alpha^2} = \frac{25}{k^2} - 2$$

Question 5

a) (i) $\ddot{x} = \frac{d}{dx} \left(\frac{v^2}{2} \right)$

$$= \frac{d}{dx} (24 + 8x - 2x^2) \quad \boxed{\checkmark}$$

$$= 8 - 4x$$

$$= -4(x-2) \quad \boxed{\checkmark}$$

$\therefore$ particle executes SHM centre at $x=2$

(ii) $v^2 \geq 0$

$$\therefore 48 + 16x - 4x^2 \geq 0$$

$$x^2 - 4x - 12 \leq 0$$

$$(x+2)(x-6) \leq 0$$

$$-2 \leq x \leq 6$$

$$\therefore \text{Amplitude} = 4 \quad \boxed{\checkmark}$$

(iii) Since $\ddot{x} = -4(x-2)$ from (i)

$$\therefore n = 2$$

Let the expression for displacement be

$$x = 4 \sin(2t + \alpha) + 2 \quad \boxed{\checkmark}$$

$$x = 6 \text{ when } t = 0$$

$$6 = 4 \sin \alpha + 2$$

$$4 = 4 \sin \alpha$$

$$\alpha = \frac{\pi}{2}$$

$$\therefore x = 4 \sin\left(2t + \frac{\pi}{2}\right) + 2$$

$$= 4 \cos 2t + 2 \quad \boxed{\checkmark}$$

Alt 1 Assume $x = 4 \cos(2t + \theta) + 2$ $\boxed{\checkmark}$

$$x = 6 \text{ when } t = 0$$

$$6 = 4 \cos \theta + 2$$

$$4 = 4 \cos \theta$$

$$\theta = 0$$

$$\therefore x = 4 \cos 2t + 2 \quad \boxed{\checkmark}$$

Alt 2.

$$v^2 = 48 + 16x - 4x^2$$

$$v = 2\sqrt{12 + 4x - x^2}$$

$$\frac{dx}{dt} = 2\sqrt{12 + 4x - x^2}$$

$$\int \frac{dx}{\sqrt{12 + 4x - x^2}} = \int 2 dt$$

$$\int \frac{dx}{\sqrt{16 - (x-2)^2}} = 2t + C$$

$$\sin^{-1} \frac{x-2}{4} = 2t + C$$

$$\frac{x-2}{4} = \sin(2t + C)$$

$$\therefore x = 6 \text{ when } t = 0$$

$$\frac{6-2}{4} = \sin C$$

$$1 = \sin C$$

$$\therefore C = \frac{\pi}{2}$$

$$\therefore \frac{x-2}{4} = \sin(2t + \frac{\pi}{2})$$

$$x = 4 \sin(2t + \frac{\pi}{2}) + 2$$

$$= 4 \cos 2t + 2$$

$$b) (i) \quad \dot{y} = -gt + \frac{\sqrt{3}V}{2}$$

$$\therefore y = -\frac{1}{2}gt^2 + \frac{\sqrt{3}Vt}{2} + C$$

$$y = 0 \text{ when } t = 0$$

$$\therefore C = 0$$

$$\therefore y = -\frac{1}{2}gt^2 + \frac{\sqrt{3}Vt}{2}$$

(1)

$$(ii) \quad x = \frac{Vt}{2}$$

$$\therefore t = \frac{2x}{V}$$

(2)

Put (2) into (1)

$$y = -\frac{1}{2}g\left(\frac{2x}{V}\right)^2 + \frac{\sqrt{3}V}{2}\left(\frac{2x}{V}\right)$$

$$= -\frac{2gx^2}{V^2} + \sqrt{3}x$$

$$\therefore y = \sqrt{3}x - \frac{2gx^2}{V^2}$$

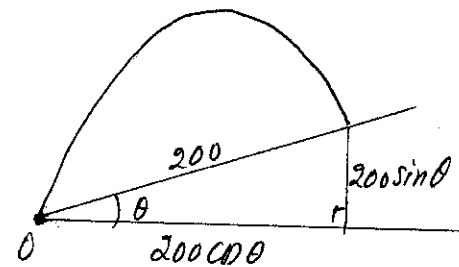
(3)

iii) The point where the arrow hits the hillside is

$$(200 \cos \theta, 200 \sin \theta)$$

Substitute into (3)

$$200 \sin \theta = \sqrt{3}(200 \cos \theta) - \frac{2g}{V^2}(200 \cos \theta)^2$$



$$200 \sin \theta = 200\sqrt{3} \cos \theta - \frac{80000g \cos^2 \theta}{V^2}$$

$$\sin \theta = \sqrt{3} \cos \theta - \frac{400g \cos^2 \theta}{V^2}$$

$$\frac{\sin \theta}{\cos \theta} = \sqrt{3} - \frac{400g \cos \theta}{V^2}$$

$$\text{ie } \tan \theta = \sqrt{3} - \frac{400g \cos \theta}{V^2} \quad (4) \quad \checkmark$$

(iv) Similarly, the pt where the arrow hits the downward slope is $(300 \cos \theta, -300 \sin \theta)$

$$\therefore -300 \sin \theta = \sqrt{3}(300 \cos \theta) - \frac{2g}{V^2}(300 \cos \theta)^2$$

$$-\frac{\sin \theta}{\cos \theta} = \sqrt{3} - \frac{600g \cos \theta}{V^2}$$

$$\text{ie } \tan \theta = \frac{600g \cos \theta}{V^2} - \sqrt{3} \quad (5) \quad \checkmark$$

$$(4) \times 3 \quad 3 \tan \theta = 3\sqrt{3} - \frac{1200g \cos \theta}{V^2} \quad (6) \quad \checkmark$$

$$(5) \times 2 \quad 2 \tan \theta = \frac{1200g \cos \theta}{V^2} - 2\sqrt{3} \quad (7) \quad \checkmark$$

$$(6) + (7) \quad 5 \tan \theta = \sqrt{3}$$

$$\therefore \tan \theta = \frac{\sqrt{3}}{5} \quad \checkmark$$

$$\theta = \tan^{-1} \frac{\sqrt{3}}{5}$$

Question 6

a) (i) y-coordinate of point B = y-coord. of pt A

and given $\sin x = y$, the other solution to this equation is $\pi - x$ since $\sin(\pi - x) = \sin x$, \checkmark

$\therefore$ the coordinates of pt B are $(\pi - x, \sin x)$

$$(ii) \quad AB = (\pi - x) - x$$

$$= \pi - 2x \quad \checkmark$$

$$AD = \sin x$$

$\therefore$ Area of ABCD is $A(x) = (\pi - 2x) \sin x$

$$(iii) \quad \frac{dA}{dx} = -2 \sin x + (\pi - 2x) \cos x$$

$$\left. \begin{aligned} \frac{d^2A}{dx^2} &= -2 \cos x - 2 \cos x - (\pi - 2x) \sin x \\ &= -4 \cos x - (\pi - 2x) \sin x \\ &< 0 \quad \text{for all } 0 < x < \frac{\pi}{2} \end{aligned} \right\} \checkmark$$

$\therefore$ A will be max when $\frac{dA}{dx} = 0$

$$\text{ie } -2 \sin x + (\pi - 2x) \cos x = 0$$

$$2 \sin x = (\pi - 2x) \cos x$$

$$2 \tan x = \pi - 2x$$

$$\text{ie } 2 \tan x + 2x - \pi = 0 \quad \checkmark$$

by result of (b) (i) of Q 2 $x = 0.72$,

$$\sin x = \sin 0.72$$

$$= 0.6594$$

$\therefore$ The dimensions of the largest rectangle are $\pi - 2(0.72)$ by 0.66 , ie 1.70×0.66 $\checkmark$

b) (i) $f(x) = \sin^{-1} \frac{1}{x} + \cos^{-1} \frac{1}{x}$

$$f'(x) = \frac{1}{\sqrt{1 - \frac{1}{x^2}}} \times \left(-\frac{1}{x^2}\right) - \frac{1}{\sqrt{1 - \frac{1}{x^2}}} \left(-\frac{1}{x^2}\right)$$

$$= 0$$

(ii) domain: $-1 \leq \frac{1}{x} \leq 1$

$$x \leq -1 \text{ or } x \geq 1$$

Since $f'(x) = 0$

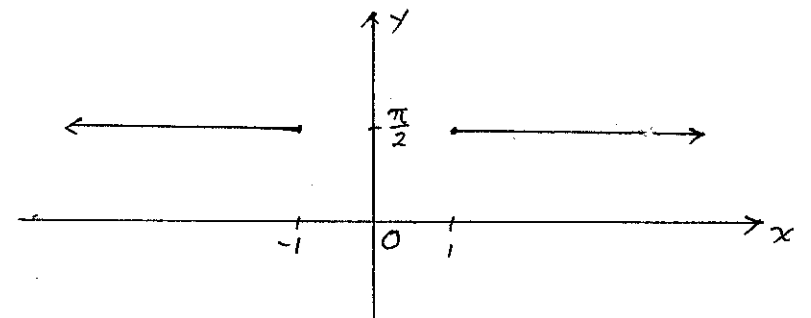
$$\therefore f(x) = c \text{ (a constant)}$$

since $f(1) = \sin^{-1} 1 + \cos^{-1} 1$
 $= \frac{\pi}{2}$

$$\begin{aligned} f(-1) &= \sin^{-1}(-1) + \cos^{-1}(-1) \\ &= -\sin^{-1} 1 + \pi - \cos^{-1} 1 \\ &= -\frac{\pi}{2} + \pi \\ &= \frac{\pi}{2} \end{aligned}$$

$$\therefore f(x) = \frac{\pi}{2}$$

$$x \leq -1 \text{ or } x \geq 1$$



(iii) $y = f(x)$ and $y = f^{-1}(x)$, if intersect, must intersect on the line $y = x$.

$$\text{As } f(x) = \frac{\pi}{2},$$

$$\therefore \text{ at pt of intersection } x = f(x) = \frac{\pi}{2}$$

$$\therefore \text{ pt of intersection is } \left(\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Question 7

a) (i) $BM^2 = OB^2 - OM^2$
 $= r^2 - h^2$

$$\therefore BM = \sqrt{r^2 - h^2}$$

(ii) $AB + CD = AD - BC$

$$= b - 2BM$$

$$= b - 2\sqrt{r^2 - h^2}$$

$$\therefore \text{ Area } A = k \times (AB + CD)$$

$$= k[b - 2\sqrt{r^2 - h^2}]$$

$$\begin{aligned}
 \text{(iii)} \quad \frac{dA}{dt} &= \frac{d}{dt} k[b - 2\sqrt{r^2 - h^2}] \\
 &= \frac{d}{dh} k[b - 2\sqrt{r^2 - h^2}] \times \frac{dh}{dt} \quad \boxed{\checkmark} \\
 &= \frac{-2k}{2\sqrt{r^2 - h^2}} \times -2h \times \frac{dh}{dt} \\
 &= \frac{2hk}{\sqrt{r^2 - h^2}} \frac{dh}{dt} \quad \boxed{\checkmark}
 \end{aligned}$$

when $k = 300$, $b = 200$, $r = 50$, $h = 30$
 and $\frac{dh}{dt} = -0.6 \text{ cm/day}$

$$\begin{aligned}
 \frac{dA}{dt} &= \frac{2 \times 30 \times 300}{\sqrt{50^2 - 30^2}} \times (-0.6) \text{ cm}^2/\text{day} \\
 &= -270 \text{ cm}^2/\text{day}
 \end{aligned}$$

$\therefore$ surface area is decreasing at $90 \text{ cm}^2/\text{day}$. $\boxed{\checkmark}$
 [also accept $0.009 \text{ m}^2/\text{day}$]

$$\begin{aligned}
 \text{b) (i) At P} \quad \left. \begin{aligned} x &= a \cos 2\theta \\ &= a(2\cos^2\theta - 1) \end{aligned} \right\} \boxed{\checkmark} \quad (1) \\
 y &= a \cos \theta \\
 \therefore \cos \theta &= \frac{y}{a} \quad (2)
 \end{aligned}$$

Put (2) into (1)

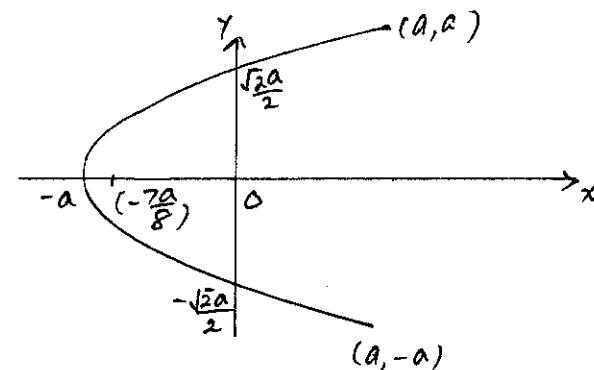
$$\left. \begin{aligned} x &= a\left(\frac{2y^2}{a^2} - 1\right) \\ &= \frac{2y^2}{a} - a \\ ax &= 2y^2 - a^2 \end{aligned} \right\} \boxed{\checkmark}$$

$$\begin{aligned}
 \text{i.e. } y^2 &= \frac{ax}{2} + \frac{a^2}{2} \\
 &= \frac{a}{2}(x+a)
 \end{aligned}$$

(ii) The focus is a parabola with vertex $(-a, 0)$
 and focal length $\frac{a}{8}$
 $\therefore$ focus is $(-\frac{7a}{8}, 0)$ $\boxed{\checkmark}$

$$\begin{aligned}
 \text{since } x &= a \cos 2\theta \\
 \text{and } -1 &\leq \cos 2\theta \leq 1 \\
 \therefore -a &\leq x \leq a
 \end{aligned}$$

Similarly $-a \leq y \leq a$ $\boxed{\checkmark}$



$\boxed{\checkmark}$ for slope
 a some
 indication
 of scale

(iii)

$$x = a \cos 2\theta$$

$$\frac{dx}{d\theta} = -2a \sin 2\theta$$

$$y = a \cos \theta$$

$$\frac{dy}{d\theta} = -a \sin \theta$$

$$\therefore \frac{dy}{dx} = \frac{-a \sin \theta}{-2a \sin 2\theta}$$

$$= \frac{\sin \theta}{2 \sin 2\theta}$$

$$= \frac{1}{4 \cos \theta}$$

$$\therefore \text{at } \theta = \frac{\pi}{3}$$

$$\frac{dy}{dx} = \frac{1}{4 \cos \frac{\pi}{3}}$$

$$= \frac{1}{2}$$



Alternatively: $y^2 = \frac{a}{2}(x+a)$

$$y = \sqrt{\frac{a(x+a)}{2}}$$

$$\frac{dy}{dx} = \sqrt{\frac{a}{2}} \cdot \frac{1}{2\sqrt{x+a}}$$

$$= \frac{1}{2} \sqrt{\frac{a}{2(x+a)}}$$

when $\theta = \frac{\pi}{3}$, $x = a \cos \frac{2\pi}{3}$

$$= -\frac{a}{2}$$

$$y = a \cos \frac{\pi}{3}$$

$$= \frac{a}{2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} \sqrt{\frac{a}{2(-\frac{a}{2}+a)}}$$

$$= \frac{1}{2}$$



At $\theta = \frac{\pi}{3}$, $x = a \cos \frac{2\pi}{3}$

$$= -\frac{a}{2}$$

$$y = a \cos \frac{\pi}{3}$$

$$= \frac{a}{2}$$

$\therefore$ Equation of the tangent is

$$y - \frac{a}{2} = \frac{1}{2} \left(x + \frac{a}{2} \right)$$

$$2y - a = x + \frac{a}{2}$$

$$4y - 2a = 2x + a$$

$$\text{or } 2x - 4y + 3a = 0$$

